Supplementary Material for "Nodal Frequency-Secure Real-Time Dispatch of Inverter-Dominated Power Systems With Learned Constraints Monotone in Activatable Reserve"

This document details the nodal post-contingency model of Section II-A and the training of the PICNN of Section III-B of the paper. Equations and tables numbered with the prefix S belong to this document, and all other numbers refer to the paper.

I. Nodal Post-Contingency Model

The model has the structure of the nodal frequency response models used in frequency-constrained scheduling and nodal frequency analysis [1]–[4]. Synchronous generators (SGs) and grid-forming (GFM) inverter-based resources (IBRs) are swing devices with (virtual) inertia, damping, and a power–frequency control loop. Grid-following (GFL) IBRs and battery energy storage system (BESS) units inject their frequency response at their buses. All devices are coupled through the network susceptances, and the bus frequencies follow from the device frequencies by the frequency divider [5]. Every unit is represented individually, and its response is limited by its reserve in the dispatch. Powers are in p.u. on the base P^B unless stated otherwise.

A. Devices

For contingency c, let $\mathcal{S}_c$ be the surviving swing devices, which are the SGs with positive active-power capability and the GFM IBRs online at the penetration level, and let $\mathcal{F}_c$ be the surviving GFL IBRs and BESS units. A GFM IBR is represented, like an SG, by a swing device with its virtual inertia [2]. Swing device u at bus $n(u)$, with rating S_u, inertia constant H_u on its rating (H^{V} for a GFM IBR), and damping D_u, obeys

$$\dot{\delta}_u = 2\pi f_0(\omega_u - 1), \quad 2M_u\dot{\omega}_u = p^{\mathrm{m}}_u - p^{\mathrm{e}}_u - D_u(\omega_u - 1), \tag{S1}$$

where $M_u = H_u S_u / P^B$, δ_u and ω_u are its internal angle and per-unit speed, and p^{m}_u and p^{e}_u are its mechanical or control power and its electrical power. Each swing device follows the droop setpoint

$$g_u = \mathrm{sat}_{[\underline{p}_u, \overline{p}_u]}\Big(P_u + \frac{S_u}{0.05P^B}(1 - \omega_u)\Big), \tag{S2}$$

where P_u and R_u are its output and reserve in the dispatch, $\overline{p}_u = P_u + R_u$, $\underline{p}_u = \max(P_u - R_u, 0)$, and $\mathrm{sat}_{[a,b]}$ clips its argument to $[a, b]$. The droop is 5% on the rating. The governor and turbine of an SG have the structure of the TGOV1 model [6], a valve lag T_1 followed by a lead–lag reheat stage (T_2, T_3):

$$\begin{aligned} &T_1\dot{x}_{1,u} = g_u - x_{1,u}, \quad T_3\dot{x}_{2,u} = (1-a)\,x_{1,u} - x_{2,u}, \\ &p^{\mathrm{m}}_u = a\,x_{1,u} + x_{2,u}, \quad a = T_2/T_3. \end{aligned} \tag{S3}$$

A GFM IBR delivers its setpoint through a first-order activation lag [4],

$$T^{\mathrm{A}}\dot{p}^{\mathrm{m}}_u = g_u - p^{\mathrm{m}}_u. \tag{S4}$$

The units in $\mathcal{F}_c$ respond to the center-of-inertia (COI) frequency

$$f^{\mathrm{coi}} = f_0\Big[1 + \sum_{u\in\mathcal{S}_c} M_u(\omega_u - 1)\Big/\sum_{u\in\mathcal{S}_c} M_u\Big] \tag{S5}$$

through a first-order lag,

$$\tau_j\dot{p}^{\mathrm{F}}_j = \mathrm{sat}_{[0,\overline{r}_j]}\Big(\frac{S_j}{0.05P^B}\Big(1 - \frac{f^{\mathrm{coi}}}{f_0}\Big)\Big) - p^{\mathrm{F}}_j, \quad j \in \mathcal{F}_c, \tag{S6}$$

where p^{F}_j is the incremental injection of unit j and S_j its rating. A GFL IBR i has $\overline{r}_j = R_i$ and $\tau_j = T^{\mathrm{A}}$, and a BESS unit b has $\overline{r}_j = r^{\mathrm{F}}_b/P^B$ and $\tau_j = 0.1$ s.

B. Network and Bus Frequencies

Let $\boldsymbol{B}^{\mathrm{net}}$ be the susceptance Laplacian of the in-service branches, in which branch (n, m) with reactance x_{nm} adds $1/x_{nm}$ to the entries (n, n) and (m, m) and $-1/x_{nm}$ to the entries (n, m) and (m, n). The internal node of swing device u is coupled to bus $n(u)$ through $x_u = X_u P^B/S_u$, where X_u is the transient reactance of an SG or the coupling reactance of a GFM IBR on its rating. With $\boldsymbol{X} = \mathrm{diag}(x_u)$ and the bus incidence matrices $\boldsymbol{E}_{\mathrm{S}}$ of the swing devices and $\boldsymbol{E}_{\mathrm{F}}$ of the units in $\mathcal{F}_c$, let

$$\boldsymbol{B}_{\mathrm{BB}} = \boldsymbol{B}^{\mathrm{net}} + \boldsymbol{E}_{\mathrm{S}}\boldsymbol{X}^{-1}\boldsymbol{E}_{\mathrm{S}}^{\mathsf{T}}, \quad \boldsymbol{B}_{\mathrm{BS}} = \boldsymbol{E}_{\mathrm{S}}\boldsymbol{X}^{-1}. \tag{S7}$$

The bus angles $\boldsymbol{\theta}$ and the electrical powers of the swing devices follow from

$$\boldsymbol{B}_{\mathrm{BB}}\boldsymbol{\theta} = \boldsymbol{B}_{\mathrm{BS}}\boldsymbol{\delta} + \boldsymbol{p}^{\mathrm{B}} + \boldsymbol{E}_{\mathrm{F}}\boldsymbol{p}^{\mathrm{F}}, \quad p^{\mathrm{e}}_u = (\delta_u - \theta_{n(u)})/x_u, \tag{S8}$$

where $\boldsymbol{p}^{\mathrm{B}}$ collects the scheduled outputs of the GFL IBRs and BESS units minus the bus demands, which are frequency-independent. The bus frequencies follow from the frequency divider [5]:

$$\boldsymbol{f} = f_0\big[\boldsymbol{1} + \boldsymbol{B}_{\mathrm{BB}}^{-1}\boldsymbol{B}_{\mathrm{BS}}(\boldsymbol{\omega} - \boldsymbol{1})\big]. \tag{S9}$$

C. Contingency Simulation and Worst-Bus Drop

The pre-contingency state is the equilibrium of the intact system at the dispatch, with $p^{\mathrm{m}}_u = p^{\mathrm{e}}_u = P_u$ for every swing device and the bus demands scaled by a common factor that allocates the network losses of the dispatch in proportion to them. At $t = 0$ the units of $\mathcal{T}_c$ trip: their internal nodes, scheduled outputs, inertia, and reserve are removed, and $\boldsymbol{B}_{\mathrm{BB}}$

TABLE SI
PARAMETERS OF THE NODAL POST-CONTINGENCY MODEL

Parameter	Value
Nominal frequency f_0	50 Hz
Power base P^B	100 MVA
SG inertia constant H_u	3.5 s on S_u
SG transient reactance X_u	0.5 p.u. on S_u
TGOV1 time constants T_1, T_2, T_3	0.05, 1.0, 2.1 s
GFM virtual inertia H^{V}	2.0 s on S_u
GFM coupling reactance X_u	0.2 p.u. on S_u
Droop of SGs, IBRs, and BESS units	5% on the rating
IBR activation time constant T^{A}	1.0 s
BESS response time constant	0.1 s
Damping D_u of SGs and GFM IBRs	2.5 p.u. on P^B
Damping D_u of IEEE 300 SGs	2.0 p.u. on S_u
UFLS relay window w	0.15 s
Simulation horizon T_{s}	12 s

TABLE SII
TRAINING SETTINGS OF THE PICNN

Setting	Value
Hidden layers L, width w	3, 64 in both branches
Activations	ReLU (convex), ELU (context)
Weight initialization	$\mathcal{N}(0, 0.05^2)$, zero biases
Sign-constrained weights	softplus, raw values from -2
Input and label scaling	standardized, training statistics
Loss	Huber, $\delta = 0.05$ (standardized)
Optimizer	AdamW, rate 10^{-3}, decay 10^{-4}
Learning-rate schedule	cosine annealing, 220 epochs
Batch size	256
Maximum epochs	220
Early stopping	validation RMSE, patience 25
Random seeds	4, best validation RMSE kept

and $\boldsymbol{B}_{\mathrm{BS}}$ are rebuilt for the surviving devices. The internal angles of the surviving devices are continuous, their speeds start at $\omega_u(0) = 1$ and their governor and activation states at the pre-contingency steady state, and $\boldsymbol{p}^{\mathrm{F}}(0) = \mathbf{0}$. The model is integrated over $T_{\mathrm{s}} = 12$ s with relative and absolute error tolerances of 10^{-7} and 10^{-9}. For bus n, the relay-window drop is

$$\Delta f_{n,c}^{\mathrm{nad}}(x) = \max_{0 \le t \le T_{\mathrm{s}}} \min_{t-w \le s \le t} \big[f_0 - f_n(s) \big], \qquad \text{(S10)}$$

with $f_n(s) = f_0$ for $s < 0$ and the UFLS relay window $w = 0.15$ s, and the worst-bus drop $\Delta f_c^{\mathrm{wb}}(x) = \max_{n \in \mathcal{N}} \Delta f_{n,c}^{\mathrm{nad}}(x)$ enters (1s). Table SI lists the parameters, and the unit ratings and network data are in the accompanying test-system data.

II. PICNN TRAINING

The PICNN of Section III-B learns the worst-bus drop of the model of Section I, one label per trip at a labeled operating point. The labeled operating points of each penetration level are split at random into training, validation, and calibration sets, so that all trips of an operating point fall in the same set. On IEEE 118, these sets hold 12,528, 1,728, and 3,888 rows, and the calibration rows form the calibration sets of Section IV-A. Table SII lists the training settings. The network is trained from four random seeds, and the one with the lowest validation error is deployed. The IEEE 300 surrogate and the surrogate of the 51% fleet of Section V-E are trained with the same settings on labels of their own systems.

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